

Quotients

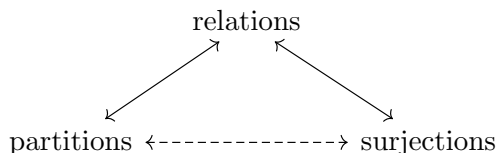
A surjection $p : X \rightarrow Z$ induces an equivalence relation $\sim$ on X , and p factors through the quotient map:

$$\begin{array}{ccc} X & \xrightarrow{q} & X/\sim \\ & \searrow p & \downarrow \cong \bar{p} \\ & & Z \end{array}$$

Universal property. For every function $f : X \rightarrow Y$ that is constant on equivalence classes there is exactly one $\bar{f}$ with $f = \bar{f} \circ q$:

$$\begin{array}{ccc} X & \xrightarrow{q} & X/\sim \\ & \searrow f & \downarrow \exists! \bar{f} \\ & & Y \end{array}$$

The three descriptions are equivalent: equivalence relations on X , partitions of X and surjections with domain X (up to bijection of the codomain).



Exercise. Show that the composite of two quotient maps is a quotient map, and draw the square

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D \end{array}$$

for the quotients of $\mathbb{Z}$ by $4\mathbb{Z}$ and $2\mathbb{Z}$.

Products. The product $X \times Y$ comes with projections, and maps into it are pairs of maps:

$$\begin{array}{ccccc} & & W & & \\ & \swarrow f & \downarrow (f,g) & \searrow g & \\ X & \xleftarrow{\pi_1} & X \times Y & \xrightarrow{\pi_2} & Y \end{array}$$

Text after the diagram fills the box so that it continues to the next page if the page is short: the frame is then measured and drawn for each part. A box split across pages is a framed environment's hardest case, and the one **framed** measures most often.

Coproducts. Dually, maps out of a disjoint union are pairs of maps.

$$\begin{array}{ccccc} X & \xrightarrow{\iota_1} & X \sqcup Y & \xleftarrow{\iota_2} & Y \\ & \searrow f & \downarrow & \swarrow g & \\ & & W & & \end{array}$$